THOMAS SIROKY

Riverside, CA | 951-434-1281 | thomasiroky@gmail.com | http://www.linkedin.com/in/thomas-siroky|

EDUCATION

California Baptist University, Riverside CA (ABET) 3.92 GPA

Bachelor of Science in Mechanical Engineering September 2023 – Expected May 2027

MECHANICAL ENGINEERING EXPERIENCE

Mechanical Engineer Intern | Opto 22, Temecula CA April – August 2026

Automated Induction Soldering Machine for Solid State Relays

- Designed, manufactured, and assembled an automated induction soldering machine that fed and placed four terminals onto a PCB, applied flux, transferred PCBs to a chain conveyor, and conveyed boards through a channel style induction coil.
- Designed custom mechanisms and fixtures in Autodesk Inventor using DFM/DFA principles, then manufactured aluminum and Delrin components using manual/CNC mills, 80/20 aluminum extrusions, and FDM/SLA 3D printing.
- Integrated five A6 servo drives with an Opto 22 groov EPIC over Modbus RS-485; wrote PLC control routines for homing, absolute positioning, conveyor control, sensor interlocks, and automated machine sequencing in Opto 22 PAC Control.
- Wired and commissioned the servo, pneumatic, sensor, and control systems; troubleshot mechanical and control issues until the machine could complete full automated cycles with 100-board feeder capacity.

SELECT ENGINEERING PROJECTS

ACE (Autonomous Court Enhancer) April 2026

- Led a six-person team in the end-to-end design and fabrication of an autonomous hardwood court-cleaning robot, integrating mechanical, electrical, and embedded control systems into a fully functional platform.
- Applied SolidWorks modeling, FEA/CFD, and engineering calculations to drive design decisions; developed an Arduino-based 12 V control system with geared DC motors, motor drivers, pump control, and PID ultrasonic sensor feedback.

Reverse Engineering & Redesign of External Gear Pump December 2025

- Led a four-person team in the disassembly and characterization of a plastic external gear pump to analyze functionality, capturing dimensional geometry using a FARO arm and conventional metrology, noting design flaws for improvement.
- Analyzed and redesigned the pump to create higher flow capabilities in SolidWorks. Manufactured and assembled using 3D-printed PLA. Achieved and sustained a flow rate at 54% of the original pump, identifying material-related vacuum limitations associated with PLA.

Titan Building Blocks June 2023

- Designed and modeled mechanical components in SolidWorks (CAD), programmed toolpaths in Mastercam (CAM), and manufactured parts on a HAAS 3-axis CNC mill.
- Gained experience in the design-to-manufacture process, applying design for manufacturing (DFM) principles to improve efficiency and precision.

TECHNICAL SKILLS

- **CAD Design:** SolidWorks, Autodesk Inventor, AutoCAD; part modeling, assemblies, engineering drawings, DFM/DFA, and mechanism design.
- **Analysis:** SolidWorks and Inventor FEA/CFD, mechanical design calculations, tolerance-based design.
- **CAM/CNC:** Mastercam and Fusion 360 CAM; 3-axis toolpath generation and HAAS CNC mill operation.
- **Manual Manufacturing:** Manual milling, drilling, tapping, fitting; fabrication of aluminum/steel/engineering plastics.
- **Electromechanical Systems:** Servo/DC/Stepper motors, linear actuators, pneumatics, pumps, control panel wiring.
- **Industrial Controls:** Opto 22 PAC Control, groov EPIC, Modbus, homing, absolute positioning, machine sequencing.
- **Embedded Systems:** Arduino/Rasberry Pi, motor drivers, PWM control, sensor integration, circuit prototyping.